

Ganit

data speaks

NovaPay – Enterprise Data Platform

Case Study

Unified Digital Payments Analytics on Azure with
Real-Time Fraud Detection & Regulatory Compliance

BFSI | Digital Payments

India & SEA Operations

Unifying Digital Payments, Fraud & Compliance Data for NovaPay with a Scalable Azure Data Platform and Real-Time Analytics Across India & SEA

Problem Statement

- Siloed UPI / card / wallet transaction data
- Reactive fraud detection (rule-based, T+1)
- Manual PCI-DSS & RBI compliance reporting
- No real-time merchant settlement view

Solution Proposed by Ganit

NovaPay needed a unified analytics platform ingesting UPI rails, card networks, and wallet systems — enabling real-time fraud scoring and automated RBI / PCI-DSS compliance reporting across India & SEA.

- ✓ **Unified Azure Data Lake + Synapse DWH**
- ✓ **Governance via Purview & Azure Policy**

Architecture

UPI Rails / Card Networks / Wallets / AML Feeds

Azure Data Factory + Databricks (Streaming + Batch)

ADLS Gen2 (Raw / Cleansed / Curated)

Azure Synapse Analytics (Payments DWH)

Power BI — Fraud & Compliance Dashboards

Business Value Achieved

✓ Unified fraud view across India & SEA

✓ Automated PCI-DSS / RBI reporting

✓ Migrated 6 TB of payments history

✓ Fraud detection from T+1 to real-time

✓ Power BI dashboards for settlements

✓ Purview governance cutting incidents 80%

Ganit unified NovaPay's payments, fraud & compliance data on Azure, enabling real-time fraud scoring and automated regulatory reporting across India & SEA

Delivered a cloud-native Azure Data Platform ingesting data from UPI payment rails (NPCI), card networks (Visa/Mastercard), and digital wallets via Azure Data Factory and Databricks streaming pipelines — governed by Microsoft Purview — and surfaced via Power BI Premium for real-time fraud and compliance analytics across India & Southeast Asia.

Regulatory Compliance

Automated PCI-DSS Level 1 and RBI Payment Aggregator reporting, eliminating manual reconciliation and reducing submission errors by 88% across all regulatory jurisdictions.

Cost Optimisation

Replaced 4 legacy on-premise data warehouses and batch systems with a unified Azure platform, achieving ~84% reduction in infrastructure costs while enabling elastic scaling for festival-peak transaction volumes.

Risk & Resilience

Built a multi-zone ADLS Gen2 data lake (Raw/Cleansed/Curated) with Purview lineage handling 6+ TB of payments history, supporting 250+ concurrent Power BI users including risk, ops, and compliance teams.

Business Growth

Enabled real-time ML fraud scoring on transaction streams via Databricks, reducing false positives by 67% and giving merchant success and treasury teams instant visibility into settlement cycles and chargeback ratios.

Siloed payments data, reactive fraud detection, and manual compliance submissions blocked unified financial analytics across NovaPay's digital payments operations

NovaPay operates across India and Southeast Asia as a full-stack digital payments platform — processing UPI transactions (NPCI), card payments (Visa/Mastercard/RuPay), and proprietary digital wallet settlements. With over 120 million active users and 3 billion+ annual transactions, NovaPay processes payments across retail, merchant, B2B, and cross-border corridors.

Transaction data was siloed across payment rails, fraud signals were rule-based and retrospective (T+1), and compliance reporting to RBI and PCI-DSS was manually assembled from CSV exports. This created systemic risk in fraud exposure, regulatory non-compliance, and inability to provide merchants with real-time settlement transparency.

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Challenge 1 — Fragmented Payments Data

UPI, card, wallet & cross-border transaction data siloed across NPCI, Visa/MC, and proprietary systems — no unified ingestion pipeline or consolidated ledger.

2

Challenge 2 — Reactive Fraud Detection

Rule-based fraud engine operating on T+1 batch scores, missing real-time transaction anomalies — resulting in high chargeback ratios and false-positive merchant blocks.

3

Challenge 3 — Manual Regulatory Reporting

PCI-DSS Level 1 and RBI Payment Aggregator compliance reports manually assembled from CSV exports — causing submission delays, version mismatches, and reconciliation errors.

4

Challenge 4 — No Merchant Settlement Visibility

Merchants lacked real-time settlement dashboards; treasury had no intraday liquidity view — leading to settlement disputes and trust erosion with enterprise merchants.

Solution Overview

Robust & Scalable Architecture

- › UPI (NPCI) / card (Visa/MC/RuPay) / wallet data ingested via ADF with CDC & delta loads.
- › Fraud signals & transaction streams via Azure Event Hubs + Databricks ML scoring.
- › Regulatory feeds (RBI, PCI-DSS DSS audit logs) processed with schema validation.
- › Raw → Cleansed → Curated ADLS Gen2 zones with Purview lineage & data catalogue.

Application UI — Power BI Premium

- › Power BI Premium with DirectQuery to Synapse for real-time fraud dashboards.
- › Dashboards for Fraud Risk, Merchant Settlements, AML Alerts & Customer 360.
- › Row-level security via Azure AD groups mapped to merchant / geography.

Data Architecture

